Lóránt Nagy, PhD

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Professional Experience

Artificial Intelligence National Laboratory **2024**
Research Assistant *Budapest, Hungary*

- Worked with ALTEO on day-ahead electricity-price forecasting.
- Developed and evaluated machine-learning regression models and assessed their reliability.
- Communicated modelling choices, limitations, and findings to academic and industry collaborators.

Eötvös Loránd University, AI Research Group **2023–Present**
Research Assistant *Budapest, Hungary*

- Conduct research on machine-learning methods for stochastic processes and long-memory time series.
- Conducted an interpretability study of neural-network-based Hurst estimation, providing evidence that the model relies primarily on spectral rather than geometric information.
- Develop reinforcement-learning methods for trading under market frictions, comparing learned performance with theoretical asymptotic behaviour.

Alfréd Rényi Institute of Mathematics **2019–Present**
Research Assistant *Budapest, Hungary*

- More recent work focuses on generative AI and deep learning, particularly diffusion-based generative models.
- Developed and presented a framework for studying superlinearly mean-reverting diffusions as alternative forward processes in generative modelling, with ongoing numerical experiments comparing them with classical Ornstein–Uhlenbeck dynamics.
- Conduct research in mathematical finance, stochastic control, market frictions, and optimal investment.

Morgan Stanley & Co. LLC **2017–2018**
Interest Rate Modelling Intern *Budapest, Hungary*

- Applied quantitative methods to problems in interest-rate modelling.
- Gained experience with mathematical modelling and numerical analysis in an industry setting, as well as communicating technical work within a professional team.

Education

Central European University **2018–2022**
PhD in Mathematics *Budapest, Hungary*

Budapest University of Technology and Economics **2015–2018**
MSc in Mathematics *Budapest, Hungary*

Technical Skills

Programming and development tools: Python, PyTorch, Git, Docker, GitHub Actions, Wolfram Mathematica, LLM-assisted development workflows

Quantitative methods: stochastic modelling, simulation, optimization, statistical analysis, and mathematical finance

Cloud and deployment: Azure Static Web Apps, Azure Container Apps; basic CI/CD workflows and containerized application deployment

Machine learning tools and workflows: Weights & Biases, TensorBoard, reproducible experiment workflows, model comparison, and evaluation

Machine learning and data analysis: deep learning, neural networks, diffusion-based generative modelling, regression, and time-series forecasting

Publications

Rásonyi, M. and Nagy, L. (2021). *Optimal long-term investment in illiquid markets when prices have negative memory.* Electronic Communications in Probability, 26.

Guasoni, P., Nagy, L., and Rásonyi, M. (2021). *Young, timid, and risk takers.* Mathematical Finance, 31, 1332–1356.

Boros, D., Csanády, B., Ivkovic, I., Nagy, L., Lukács, A., and Márkus, L. (2024). *Deep learning the Hurst parameter of linear fractional processes and assessing its reliability.* Quality and Reliability Engineering International, 40, 4228–4246.

Csanády, B., Nagy, L., Boros, D., Ivkovic, I., Kovács, D., Tóth-Lakits, D., Márkus, L., and Lukács, A. (2024). *Parameter estimation of long-memory stochastic processes with deep neural networks.* Proceedings of ECAI 2024.

Rásonyi, M. and Nagy, L. (2025). *Optimal investment with transient price impact and stochastic spread.* Preprint, arXiv:2511.12093.

Rásonyi, M. and Nagy, L. *On arbitrage with linear and superlinear friction.* Manuscript in preparation.

Conference and Seminar Presentations

2026 *On the utility problem in a market where price impact is transient*
XIII Bachelier World Congress, Bologna

2025 *Non-convex optimal investment with transient price impact*
Vienna Congress on Mathematical Finance, TU Wien

2024 *Optimal execution in superlinearly mean-reverting environments*
AMK Seminar, Bolyai Institute, University of Szeged

2023 *Risk-averse trading*
ELTE–Rényi AI Workshop, Eötvös Loránd University

2023 *Random walks in random environments*
Probability Seminar, Alfréd Rényi Institute of Mathematics

2022 *Young, Timid, and Risk Takers*
Financial Mathematics Seminar, University of Michigan

2021 *On sub-Gaussian estimators of the mean*
Probability Seminar, Alfréd Rényi Institute of Mathematics

2020 *Investment in mean-reverting markets*
PhD Seminar, Central European University

2020 *How to invest in mean-reverting markets?*
Probability Seminar, Alfréd Rényi Institute of Mathematics

Additional Experience and Distinctions

2025 Intermediate-level Qualification in Copyright Law, Hungarian Intellectual Property Office.

2022 Award for Advanced Doctoral Students.

2019 Instructor, *Fundamentals of Stochastic Analysis*, Central European University.

2016 Second Place, TDK competition, *Examples in the theory of continuous trading with friction.*